Bayesian Marketing Mix Prototype

Goal: learn how marketing affects synthetic sales and brand, then forecast and plan budgets.

Built and tested: 156 weeks x 4 products x 3 price tiers (1,872 observations); 130 training weeks and 26 held-out weeks. Baseline, stockout-only, funnel-only, and combined variants passed predefined gates; 19 deterministic regression tests passed.

Methods implemented: Bayesian partial pooling (shrinkage), ad carryover, search funnel, Kalman-filtered brand scores, and profit/CVaR budget optimization.

MODEL: WHAT IT REPRESENTS

Demand and hierarchy. Product, tier, and product x tier baseline effects partially pool. Controls: price, macro, base-price growth, promotion, post-promotion dip, PrimeDays, and annual seasonality. No random-walk baseline.

Media shrinkage. Positive product responses partially pool toward learned channel centers. Display, Video, Social, and CTV have estimated adstock and saturation; paid clicks have no adstock. Social is S-shaped. Tier slopes and transforms remain shared; no-pooling superiority is untested.

Funnel. CTV, organic demand, and controls drive branded search. Search has log-Normal measurement noise; clicks are Binomial with spend-dependent capture. Sales use latent mean clicks, not observed holdout outcomes; CTV also has a direct effect.

Kalman brand state. A jointly estimated 0-100 score follows a linear-Gaussian logit state process with shocks and noisy score measurements. A Kalman filter updates states from training scores; nonlinear sales are not Kalman observations.

New business inputs. Sponsorship has impressions and separate spend at a synthetic $18 CPM. Weekly product distribution is the volume-weighted % of available stores/market carrying the product, not store count. It is fixed in planning; no separate SKU-breadth variable.

Sales and decisions. Censored log-Normal sales = min(demand, inventory). Six spend lines include sponsorship. Compare units, expected contribution-profit, and 50/50 expected/5%-CVaR objectives over 13 weeks. Margin rates are illustrative, not observed costs; budget and product proportions are fixed. CTV attribution avoids double counting.

COMBINED RUN: EVIDENCE

Sampling reliability
R-hat 1.0055; 0 divergences

Holdout log-sales error
87.3% lower RMSE*

90% predictive coverage
Demand 90.1%; sales 92.6%

Funnel coverage / CTV
Search 91.3%; clicks 94.2%; CTV lift error 2.8%

Known media-effect recovery
20/20 product intervals; max error 4.39%

Brand score holdout
RMSE 1.26 points; 87.5% coverage

Persistence / baseline demand
0.904 weekly; +8.5% per +10 prior score pts

5% profit CVaR
$944.3m baseline; $952.8m risk allocation

13-week decisions
$7.48m; units +0.54%; profit +$5.1m

*Versus a product-tier mean log-sales benchmark. All gates passed. Budget conserved; unit-objective shortfall is below 0.0001% versus the known-truth best found under the same constraints. Six local starts are not a global-optimum proof.

NEXT STEPS: MMM BEST PRACTICE AND QUANT RESEARCH

1. Identification. Use geo/lift evidence; address spend endogeneity, mediation confounding, brand construct validity, and physical distribution effects.

2. Robustness. Repeat seeds, untouched/rolling holdouts, simulation-based calibration, prior sensitivity, and misspecification stress tests.

3. Real-data pilots. Audit score surveys, exposures, cost, sales, and weighted distribution; specify plausible future demand, supply, and coverage scenarios.

4. Quant decisions. Add profit and posterior tail-loss (CVaR) constraints; research longer-horizon brand value, pacing, uncertain costs, and replenishment.

5. Dynamic models. Test richer brand shocks, nonlinear state filters, time-varying response and lag forms, and joint brand/supply dynamics; require decision value.

Boundary: one deliberately informative matched synthetic seed; not independent real-data evidence. Future inputs/coverage/inventory and margins are supplied assumptions. Kalman filtering uses scores, not nonlinear sales observations. No direct distribution constraint or separate assortment breadth. Attribution is a convention; 13 weeks omit later brand value. **Not real-world causal validation or deployment advice.**

Alternative Methods: Compare and Contrast

A frequentist Ridge model is a useful benchmark, but it is not a drop-in equivalent to this Bayesian hierarchical MMM.

BAYESIAN HIERARCHICAL MMM

What it means. Parameters are represented by probability distributions updated with observed data and prior assumptions. Posterior draws describe uncertainty about effects and predictions, conditional on the model and data.

What this prototype adds. Product effects partially pool toward shared channel centers. The model also represents nonlinear media response, a search/click funnel, censored sales, persistent brand state, and uncertainty for downstream decisions.

Costs and caveats. MCMC takes longer and requires convergence diagnostics. Results depend on model structure and priors; posterior uncertainty does not by itself establish causal effects or protect against misspecification.

FREQUENTIST RIDGE REGRESSION

What it means. Coefficients are treated as fixed but unknown. Inference asks how estimates or procedures would vary across repeated samples. Ridge fits coefficients with an L2 penalty, usually tuned by time-aware validation.

What it buys. Fast fitting, stable estimates under correlated predictors, and a simple predictive benchmark. It can show whether the more elaborate model materially improves held-out forecasts.

What it gives up. Ordinary Ridge shrinks toward zero, not toward a learned population effect; hierarchy needs extra structure. It gives point estimates, so uncertainty needs methods such as bootstrap. Standard Ridge also allows negative coefficients unless constrained.

AT A GLANCE

	Bayesian model	Ridge
Pooling	Bayesian model: explicit partial pooling across products.	Ridge: global shrinkage; group-specific pooling must be designed.
Uncertainty	Bayesian model: posterior and predictive scenarios feed profit/CVaR.	Ridge: bootstrap intervals and financial scenarios need additional work.
Model structure	Bayesian model: supports the prototype's funnel, censoring, and brand state.	Ridge: nonlinear features can be engineered, but these mechanisms need separate treatment.
Best fit	Bayesian model: richer inference and uncertainty-aware decisions.	Ridge: rapid baseline, prediction benchmark, and simpler repeated refits.

A FAIR TEST FOR THIS MMM

Fit Ridge to the same training window and inputs, using time-based cross-validation to choose its penalty. Keep all feature construction and scaling training-only; compare both models on the same untouched holdout for forecast error and calibration. Then compare attribution and optimized budgets: similar RMSE alone does not mean the decisions agree.

Practical recommendation: use the Bayesian model to compare unit, expected-profit, and CVaR allocations; replace illustrative margins with real costs before business use, and add Ridge as a fast predictive benchmark. In a linear Gaussian model with fixed Gaussian priors, Ridge corresponds to a Bayesian MAP estimate, but not the full posterior uncertainty or this prototype's hierarchical structure.

Data Contract and Identification

Start with stable entities, explicit measurement definitions, and strict time alignment. The synthetic panel is unusually clean; a real-data implementation should treat data quality and identification as first-class model components.

CURRENT ANALYSIS GRAIN

Sales are weekly at product x price-tier grain: 156 weeks, four products, three tiers. The current fit uses the first 130 weeks and reserves the final 26 weeks for a conditional holdout. Brand scores, sponsorship, distribution, search, and clicks are product-week quantities and must not be duplicated across the three price tiers.

The current generator carries sales, list price, discount, nominal/base price, macro and event controls; five media exposures/spend series and reference exposures; optional search and click mediators; inventory caps; product-week brand scores; sponsorship impressions/spend; and volume-weighted product distribution. Preserve keys, units, currencies, time zones, week-ending convention, and source lineage for every field.

REAL-DATA ADDITIONS TO THE CONTRACT

Obtain a canonical product/SKU hierarchy and tier mapping; transactions or weekly units; realized net price and promotion depth; returns/cancellations; variable COGS and contribution costs; campaign-level spend, impressions, clicks, reach/frequency, vendor, channel, and flight dates; brand survey scores with respondent counts, sampling weights, wording, field dates, and uncertainty; store availability and store-volume weights; and inventory, lost sales, allocation, and replenishment records.

Document aggregation rules before modeling: media-to-week attribution, exchange-rate and invoice treatment, price net of discount, availability versus assortment, brand-score reliability, and whether product substitutions move demand between tiers. Add validation for duplicate keys, missing periods, late revisions, negative spend, impossible prices, and inconsistent stockout flags.

TEMPORAL SPLIT AND LEAKAGE CONTROLS

Choose rolling-origin backtests that mimic the decision date. Fit transformations, scalers, priors informed by data, and feature selection on training data only. Never use future sales, future survey responses, realized future media, or known generating truth to choose an allocation. Clearly label future media, prices, events, organic demand, coverage, and inventory as forecasts or scenario assumptions.

Carry adstock history and the last filtered brand state across each forecast origin. Use holdout brand measurements for evaluation only, not as state updates. Keep a final untouched period after model selection; a synthetic paired holdout is not independent evidence of real-world generalization.

IDENTIFICATION BEFORE INTERPRETATION

A well-mixing sampler cannot fix confounded media spend. Map budget decisions, targeting, and demand shocks; inspect channel correlation and time-varying omitted variables. Use geo experiments, randomized holdouts, or credible quasi-experimental variation to anchor incrementality and mediation. Define brand construct validity and distribution reach before assigning those effects a causal or financial interpretation.

Model Blocks, Priors, and Likelihoods

The current PyMC model combines pooled demand baselines, nonlinear media response, funnel mediation, censored fulfillment, and a stochastic brand state. The values below describe this synthetic reference, not recommended universal priors.

BASELINE AND PARTIAL POOLING

Log expected sales combine a global intercept; zero-sum product, tier, and product x tier contrasts; relative price, macro, base-price growth, promotion, post-promotion dip, PrimeDays, and annual sine/cosine controls; media response; and the lagged brand effect. Contrast scales use HalfNormal priors (product 0.8, tier 0.6, interaction 0.15). These encode shrinkage toward group centers while allowing supported groups to differ.

MEDIA RESPONSE AND HIERARCHY

Display, Video, Social, and CTV use estimated geometric adstock over current week plus 12 prior weeks, with normalized weights; paid-search clicks have no carryover. Hill saturation uses power 2 for Social and power 1 otherwise. Media coefficients are positive HalfNormal(0.8) log-sales effects. Carryover decays use Beta(2,2); half-saturation uses LogNormal(0,0.6).

In hierarchical-media mode, product coefficients equal a positive population coefficient times exp(zero-sum log deviation). Learned product-deviation scales use HalfNormal(0.3) priors. Product slopes are shared across tiers; transformations and funnel parameters remain shared. Only four product groups are available, so pooling sensitivity and no-pooling comparisons remain important.

SEARCH AND CLICK FUNNEL

CTV saturation, macro, seasonality, and an observed organic-demand index drive latent branded-search intensity. Search observations are rounded log-Normal; paid clicks are Binomial conditional on searches with spend-dependent capture. Zero spend gives zero clicks. Sales use structural expected clicks, not noisy observed holdout mediators. CTV receives direct and induced-click lift; paid search receives its no-CTV click lift under the chosen ordered attribution convention.

BRAND STATE AND KALMAN SCOPE

On the logit score scale, z[t] = rho*z[t-1] + (1-rho)*target[t] + process shock[t]. The target is an intercept plus product effect, brand_media_beta times channel response features, brand_sponsorship_beta times sponsorship_ratio/(1+sponsorship_ratio), and brand_distribution_beta times (distribution_fraction-0.5). Brand channel coefficients are distinct from sales media coefficients. Process/measurement scales use HalfNormal(0.05); persistence uses Beta(8,2). A Gaussian measurement links the logit state to the observed 0-100 survey score.

The Kalman recursion predicts each state from its prior filtered state, then corrects it with training survey measurements; held-out scores are not used to update it. It propagates state uncertainty forward. Candidate future media and sponsorship therefore recompute the brand path, with process shocks included in financial predictive scenarios.

Sales have a separate current-period media term using media_beta or product_media_beta on adstock/saturation and modeled paid-click features. The previous week's filtered brand score, transformed with brand_sales_beta, adds to baseline log sales. Thus media can affect sales directly now and indirectly through later brand-driven baseline; sponsorship has no immediate-sales coefficient and works through later brand state. Sales are nonlinear/censored, not Gaussian Kalman measurements; the brand sales term uses a logistic-normal moment approximation. Validate against richer nonlinear state-space alternatives.

SALES AND CENSORING LIKELIHOOD

The reference uses Normal noise on log sales (equivalently log-Normal demand), with rounded unit counts treated continuously. When inventory is enabled, observed sales are min(demand, cap) and the likelihood is right-censored at log inventory. Forecast fulfilled means use E[min(D,cap)] analytically, not min(E[D],cap). The model does not forecast supply or replenishment.

Reproducible Workflow and Validation

Use the stages below as an implementation checklist. Preserve the ordering: lock data and assumptions first, evaluate the model second, and only publish allocation recommendations after diagnostics and decision checks pass.

RUN AND PRESERVE THE REFERENCE

Current end-to-end command: `python run_mmm.py --funnel --stockouts --hierarchical-media --brand-state --target-accept 0.99 --output outputs_kalman_profit_validated`. In this workspace, invoke it with `.venv/Scripts/python.exe`. The reference uses seed 20261007, 156 weeks, 130 training weeks, 26 holdout weeks, four NUTS chains, 1,500 tune steps, and 1,000 retained draws per chain. Use a new output directory for each specification or seed.

PIPELINE

1. Generate/import and validate the panel. 2. Fit only training sales and training mediators/scores. 3. Check sampler diagnostics and posterior predictive behavior. 4. Forecast conditional on explicit future inputs. 5. Check effect recovery/calibration where truth or experiments exist. 6. Reconcile attribution. 7. Optimize against the same fixed scenarios and constraints for posterior and benchmark/oracle comparisons. 8. Export data, configuration, diagnostics, metrics, plots, allocations, and report.

REQUIRED GATES

Sampling: max R-hat < 1.01, bulk ESS >= 400, tail ESS >= 200, zero divergences, and tree depth below the configured cap. Forecast: beat the product-tier mean benchmark and achieve 85%-97% nominal 90% coverage. Media recovery: at least four of five truths covered, maximum coefficient error <= 25%, decay error <= 0.10. Attribution error <= 25%; budget conserved; unit allocation within 2% of the best known-truth solution found under the same constraints. These are lightweight synthetic gates, not sufficient production acceptance criteria.

ADDITIONAL REAL-DATA CHECKS

Add multiple independent temporal origins, geographies, product groups, and campaign periods; simulation-based calibration; prior predictive checks and prior sensitivity; posterior predictive checks by product/tier/channel and stockout regime; missingness and measurement-error sensitivity; and model misspecification stress tests. Compare Ridge and seasonal/naive baselines on identical splits. Report forecast calibration and decision stability, not RMSE alone.

ARTIFACTS AND FAILURE BEHAVIOR

Keep `run_config.json`, `metrics.json`, posterior draws, run report, sales/mediator/brand predictions, recovery tables, unit decomposition, channel contributions, response curves, and both unit and profit allocation CSVs. A failed sampler gate must stop recommendation generation. Record code version, data snapshot, feature definitions, assumptions, and random seeds with each real-data run; never silently retune against the final holdout.

Decision Layer and Real-Data Roadmap

The optimizer is a transparent scenario tool, not an autonomous budget authority. Its outputs are conditional on the response model, future inputs, economics, risk preference, and the bounds supplied by the decision owner.

CURRENT PLANNING CONTRACT

Plan 13 weeks with one constant share per channel, fixed product proportions, fixed total weekly budget based on recent spend, and each share bounded to 40%-180% of baseline. Known CPM/CPC mappings convert spend to exposure; sponsorship is a sixth line in brand mode. Historical adstock and filtered brand state carry into planning. Future prices, controls, organic demand, distribution, and inventory are supplied scenarios.

The workflow reports three objectives: expected fulfilled units; expected net contribution profit; and 50% expected profit + 50% lower-tail CVaR at 5% across 512 predictive scenarios. CVaR is the mean profit in the worst 5% of scenarios. Scenarios include posterior draws, sales noise, inventory caps, and brand shocks. Six SLSQP starts find a best feasible point, not a global-optimality certificate.

SYNTHETIC ECONOMICS: REPLACE BEFORE USE

Profit proxy = fulfilled units x list price after discount x assumed product-tier contribution margin - media/sponsorship spend. The current illustrative rates by product and Core/Premium/Value tier are: Entry-Level Desktops 18/20/16%; Gaming PCs 22/28/19%; Premium Desktops 25/30/22%; Premium Laptops 20/26/18%. Media reference costs are Display/Video/Social/CTV CPM $4/$12/$9/$22, paid-search CPC $1.50, and sponsorship CPM $18.

These are not observed unit costs. Add verified variable COGS, rebates, fulfillment, returns, taxes, and any contribution definitions approved by Finance. Model uncertainty in price, margin, media costs, supply, and forecast inputs where decision material. A fixed total spend is common to all candidate allocations; clarify whether budget spend is a hard constraint, a cost in the objective, or both in the real decision.

IMPLEMENTATION PHASES

Phase 0 - Data and economics: sign off on keys, grain, measurement definitions, time alignment, COGS/margins, inventory semantics, and future-input ownership.

Phase 1 - Baselines: build time-aware seasonal-naive and Ridge benchmarks; establish rolling-origin forecast/calibration results before the hierarchical MMM.

Phase 2 - Estimation: fit the simplest identified model first, then add adstock, pooling, censoring, funnel, and brand state one at a time. Require prior sensitivity and out-of-sample evidence for added complexity.

Phase 3 - Causal validation: reconcile media effects with experiments or credible quasi-experiments; validate brand-score construct, state persistence, distribution measurement, and mediation assumptions.

Phase 4 - Decision pilot: run unit, expected-profit, and risk-aware allocations in shadow mode. Review constraint sensitivity, CVaR level/weight, scenario coverage, profit reconciliation, and decision stability with Marketing, Finance, and Supply.

Phase 5 - Operations: version data/model/config; log recommendations and overrides; monitor drift, calibration, inventory, spend costs, and realized outcomes; define rollback and retraining rules. Promote only after agreed acceptance criteria.

OPEN LIMITATIONS AND NEXT RESEARCH

No stochastic CPM/CPC likelihood, explicit SKU breadth, distribution investment constraint, replenishment, substitution, latent organic-demand model, or long-horizon brand-value objective is fitted. The Kalman filter uses score observations; sales are not Gaussian measurement updates. Brand attribution to historical funders remains an accounting convention. One matched synthetic seed and a successful fit do not establish real-world causal validity. Prioritize experiments, independent rolling validation, richer state-space alternatives, and decision-value comparisons.